

Theorems Compulsory

Module I

- ✓ (1) First theorem of Graph Theory (or Handshaking Lemma)
For any graph G , $\sum d(v) = 2e$
- (2) The number of vertices of odd degree in a graph is even
- (3) The maximum number of edges in a graph is $\frac{n(n-1)}{2}$
- ✓ (4) A simple graph with n vertices and k components can have at most $\frac{(n-k)(n-k+1)}{2}$ edges.

Module II

- ✓ (5) A given connected graph G is Euler iff all vertices of G are of even degree
or state and prove the necessary and sufficient condition for a graph to be Euler.

6) In a complete graph with n vertices there are $(n-1)/2$ edge disjoint Hamiltonian circuits if n is an odd number.

Module III

✓ (7) Any connected graph with n vertices is a tree iff it has $(n-1)$ edges.

(Prove both sides)

(8) Every tree has either one or two centres.

Module IV

(9) The vertex connectivity in a graph cannot exceed edge connectivity.

(10) The maximum vertex connectivity is $\left\lfloor \frac{2e}{n} \right\rfloor$

✓ (11) State and prove Euler's Theorem (Euler's formula, $f = e - n + 2$)

Module V

(12) Five colour theorem - Every planar graph can be colored with 5 or less colors.